

Internship Report

AWS in an Online Education System

Introduction

Today's internship session was about Amazon Web Services (AWS). Instead of learning only the theoretical concepts of AWS services, I tried to understand how they can be applied in a real-world project. I selected an Online Education System because educational platforms need to manage student registrations, live classes, recorded video lectures, assignments, and examinations in a secure and efficient manner. AWS provides cloud services that make the system scalable, reliable, and easy to maintain.

Why Cloud Computing?

If an educational institute manages its learning platform using its own servers, it must take care of hardware, software updates, backups, storage, and security. These tasks require significant time and cost. By using AWS cloud services, the institute can focus on delivering quality education while AWS handles the infrastructure. The platform can also increase or decrease computing resources based on the number of students accessing it, especially during exam seasons, and pay only for the resources it uses.

Amazon EC2

Amazon EC2 can be used to host the Online Education System. Students, teachers, and administrators can access the platform through the EC2 server. During peak hours, such as live class sessions or online examinations, when many students access the system simultaneously, the EC2 instance can be upgraded easily without affecting the application. This ensures smooth video streaming and uninterrupted access to course content.

AWS Lambda

AWS Lambda is useful for performing automatic background tasks. For example, when a student enrolls in a course, Lambda can automatically send a confirmation email or notification. It can also remind students about upcoming live classes and assignment deadlines, notify teachers about submissions, and generate certificates automatically once a course is completed. Since Lambda runs only when required, it helps reduce operational costs.

Amazon S3

Online education platforms generate many digital files such as recorded lecture videos, study materials, presentations, assignments, and certificates. These files can be securely stored in Amazon S3. Storing content in S3 improves storage management, provides high durability, and allows authorized users to access materials whenever required. Older or less frequently accessed recordings can also be moved to lower-cost storage classes.

Database Services

Student information, course details, class schedules, assignment records, grades, and payment history can be stored in Amazon RDS. If the platform needs very fast access to frequently used data, such as live quiz results or chat messages during a class, DynamoDB can also be considered. Managed database services reduce maintenance work, improve reliability, and ensure data availability.

IAM and Security

AWS Identity and Access Management (IAM) provides secure access control for different users. Teachers can access and update course content and grades, students can view only their own courses and results, and administrators can manage the entire platform. This role-based access control protects confidential student information and improves system security.

Load Balancer and Reliability

Large online education platforms may have thousands of students accessing the system every day, especially during live classes or exam windows. An AWS Load Balancer distributes incoming requests across multiple EC2 instances to maintain smooth performance. AWS monitoring and backup services also help prevent downtime and ensure that important academic data remains safe and available even during unexpected failures.

Conclusion

From today's internship session, I understood that AWS is not only suitable for large technology companies but also for educational applications. Services such as EC2, Lambda, S3, RDS, IAM, and Load Balancer can work together to build a secure, scalable, and reliable Online Education System. This practical example helped me understand how AWS services solve real-world challenges in education instead of simply learning their definitions.